

Leonardo Herrera Vaca

State of Mexico, Mexico | lhvping@gmail.com | LinkedIn
Github | +52 5579880509

SUMMARY

Physics student and self-taught backend developer with strong experience in Python, FastAPI, PostgreSQL, and Docker. Currently building a full-stack inventory management system with JWT authentication and role-based permissions.

PROJECTS

Inventory Manager (In Progress) | Python | Postgres | FastAPI | Django | Postman

- I am building a web app for management and keeping track of items. I am building the main logic with Python using Postgres. I have tried to use both drivers (psycopg2) to connect with the database and an ORM (SQLAlchemy), and I have found that using an ORM can be beneficial at some point if a simple request is being made, but when there is a more complex operation, it is far easier and more readable just to write the SQL statement needed.
- Most of the APIs in the app are being made with FastAPI because of how convenient it is for testing, but I am still testing my API with Postman. Now turning to the operations within the web app, they consist of adding a section and being able to add items to that section, each item having a name, price, stock, ID, and the time it was created or modified. Each item can be modified, deleted, and fetched.
- I am working on developing a request for fetching data within a span of time and getting a graph of a single product or a graph of all the products. This is being developed with Pandas, and it is still in early development as well.
- Now, the use of this web app is meant for small and moderately large teams, and some of the features above should be restricted to a handful of people because if everyone could use those features, it could lead to problems or even wrong manipulation of the inventory, so only some people should be able to do certain operations. Therefore, a hierarchy system and an authentication system are needed, and for that, I am using JWT authentication, and the hierarchy system is still in early development.

TECHNOLOGIES

- **Programming Languages & Frameworks:** Python, C++, Django, FastAPI, Git.
- **Tools & Cloud:** Docker, Kubernetes, AWS, GCP, Excel.
- **Databases:** MySQL, MongoDB, Postgres.
- **Languages:** English (Professional), Spanish (Native).

EDUCATION

Bachelor's Degree In Physics	09/2024 – 09/2027
UAM-I	Mexico

Relevant Courses: Operating Systems, Databases, Experimental Physics.